

In this problem set, the questions marked with a $(\star)$ are required for students enrolled in 583 and bonus for students enrolled in 483.

1. **Schwarzschild with a Cosmological Constant** Consider Einstein's equations in vacuum, but now with a cosmological constant Λ :

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0. \quad (1)$$

- (a) (10 points) Solve for the most general, spherically symmetric metric, in coordinates (t, r, θ, ϕ) . Show that this reduces to the ordinary Schwarzschild solution when $\Lambda = 0$. You can use information from HW#5 to minimize the number of computations.
- (b) (10 points) Write down the equation of motion for radial geodesics in terms of an effective potential. Sketch the effective potential for massive particles.

2. **Metric perturbations** Consider the metric perturbation

$$h_{\alpha\beta}(x) = \begin{pmatrix} c & 0 & 0 & 0 \\ 0 & -c & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & -a \end{pmatrix} \sin k(x - t), \quad (2)$$

where c, a are arbitrary constants.

- (a) (10 points) Show how this solves the linearized Einstein equation $\delta R_{\alpha\beta} = 0$.
 - (b) (10 points) Find the functions $\xi_\alpha(x)$ that transform it into a form where $c = 0$. What is the new value of a ?
3. (20 points) $(\star)$ **Gravitational Radiation:** Two point masses, m_1 and m_2 are in a Newtonian elliptical orbit with semi-major axis a and eccentricity e . Compute da/dt and de/dt that arise due to the emission of gravitational radiation. Show that the elliptical orbit tends to be circularized, i.e. $de/dt < 0$.